Problem 1. Determine whether these statements are true or false.

a) $\emptyset \in \{\emptyset\}$
b) $\emptyset \in \{\emptyset, \{\emptyset\}\}$
c) $\{\emptyset\} \in \{\emptyset\}$
d) $\{\emptyset\} \in \{\{\emptyset\}\}$
e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$
f) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$
g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

Answer.

Problem 2. Find two sets A and B such that $A \in B$ and $A \subseteq B$.

Answer.

Problem 3. Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$.

Answer.

Problem 4. This exercise presents **Russell's paradox**. Let S be the set that contains a set x if the set x does not belong to itself, so that $S = \{x \mid x \notin x\}$.

a) Show the assumption that S is a member of S leads to a contradiction.
b) Show the assumption that S is not a member of S leads to a contradiction.

By parts (a) and (b) it follows that the set S cannot be defined as it was. This paradox can be avoided by restricting the types of elements that sets can have.

Answer.

Problem 5. Prove the idempotent laws in Table 1 by showing that

a) $A \cup A = A$
b) $A \cap A = A$

Answer.

Problem 6. Let A and B be sets. Show that

a) $(A \cap B) \subseteq A$
b) $A \subseteq (A \cup B)$
c) $A - B \subseteq A$
d) $A \cap (B - A) = \emptyset$
e) $A \cup (B - A) = A \cup B$

Answer.

Problem 7. Show that if A and B are sets with $A \subseteq B$, then

a) $A \cup B = B$
b) $A \cap B = A$

Answer.

Problem 8. Let A, B, and C be sets. Use the identity $A - B = A \cup \bar{B}$, which holds for any sets A and B, and the identities from Table 1 to show that $(A - B) \cap (B - C) \cap (A - C) = \emptyset$.

Answer.

Challenge 1. The **symmetric difference** of A and B, denoted by $A \oplus B$, is the set containing those elements in either A or B, but not in both A and B.

Determine whether the symmetric difference is associative; that is, if A, B, and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?

Answer.